

EXAMPLE 4.21

Let

$$P(s) = \frac{2s + 1}{s^2}.$$

Then by Theorem 4.19 the step response of a unity feedback system with this plant and any stabilizing controller has overshoot. If we use the 2DOF controller as in Figure 4.34 with

$$C(s) = \frac{1}{2s + 1} \begin{bmatrix} 1 & 2s + 1 \end{bmatrix},$$

then the closed-loop transfer function from $r(t)$ to $z(t)$ is

$$\frac{Z(s)}{R(s)} = \frac{1}{(s + 1)^2},$$

whose step response does not have overshoot because it is a critically damped, standard, second-order system.

The above two examples give the overwhelming advantage of the 2DOF controller over the unity feedback controller. It turns out that for any reasonable plant, it is always possible to avoid overshoot by designing a 2DOF controller.

Theorem 4.22. For any LTI plant $P(s)$ with $P(0) \neq 0$, there is a 2DOF controller $C(s)$ such that the step response from the reference $r(t)$ to the output $z(t)$ does not have overshoot.

The rest of this section is dedicated to the proof of Theorem 4.22. The proof is constructive, so it also tells how such a 2DOF controller can be designed.

Here we use the parametrization of all stabilizing 2DOF controllers given in Section 3.8. Let the plant

$$P(s) = \frac{b(s)}{a(s)}$$

be proper with $\deg a(s) = n$ and

$$C_0(s) = \frac{q(s)}{p(s)}$$

where $\deg p(s) = m$ be an initial stabilizing feedback controller. The assumption that $P(0) \neq 0$ implies that $b(0) \neq 0$. Factorize $c(s) = a(s)p(s) + b(s)q(s)$ as $c(s) = f(s)h(s)$ such that $\deg f(s) = n$ and $\deg h(s) = m$. Let

$$M(s) = \frac{a(s)}{f(s)}, \quad N(s) = \frac{b(s)}{f(s)}, \quad X(s) = \frac{p(s)}{h(s)}, \quad Y(s) = \frac{q(s)}{h(s)},$$

It follows from the theory in Section 3.8 that the set of all stabilizing 2DOF controllers is given by

$$\mathcal{T}(P) = \left\{ C(s) = \begin{bmatrix} \frac{Q_1(s)}{X(s) - N(s)Q_2(s)} & \frac{Y(s) + M(s)Q_2(s)}{X(s) - N(s)Q_2(s)} \end{bmatrix} : \right.$$

$Q_1(s)$ and $Q_2(s)$ are two arbitrary stable systems

and the closed-loop transfer function from $r(t)$ to $z(t)$ is

$$G(s) = N(s)Q_1(s) = \frac{b(s)}{f(s)}Q_1(s)$$

which is independent of $Q_2(s)$. This shows that $Q_2(s)$ has nothing to do with the step response from $r(t)$ to $z(t)$ and hence can be designed independently.

Factorize $b(s)$ as $b(s) = b_s(s)b_u(s)$ where $b_s(s)$ is a stable polynomial and $b_u(s)$ is a polynomial whose roots are the unstable roots of $b(s)$. Construct

$$Q_1(s) = \frac{f(s)}{b_s(s)b_u(0)} \prod_{i=1}^k \frac{\alpha_i}{s + \alpha_i}$$

where $\alpha_i > 0$, $i = 1, 2, \dots, k$, and the integer k is chosen so that $Q_1(s)$ is proper, i.e., $k \geq \deg f(s) - \deg b_s(s)$. With this $Q_1(s)$, we have

$$G(s) = \frac{b_u(s)}{b_u(0)} \prod_{i=1}^k \frac{\alpha_i}{s + \alpha_i}.$$

We have accomplished two objectives in designing this $Q_1(s)$. The first is to cancel the stable zeros of $N(s)$ using the stable poles of $Q_1(s)$. The second is to assign all poles of $G(s)$ to negative real numbers.

Let us first consider a number of special cases, which cover the majority of the practical cases.

Case 1: $b_u(s)$ is a constant, i.e., the plant is minimum phase

In this case,

$$G(s) = \prod_{i=1}^k \frac{\alpha_i}{(s + \alpha_i)}.$$

The statement in Problem 4.18 implies that the step response of $G(s)$ is monotonic and hence is free of overshoot.

Case 2: $b_u(s) = s - z$ with $z > 0$, i.e., the plant only has one unstable zero z that is real and positive.

In this case,

$$G(s) = \frac{\alpha_1(s - z)}{-z(s + \alpha_1)} \prod_{i=2}^k \frac{\alpha_i}{s + \alpha_i}.$$

The conclusion in Problem 4.20.2 implies that $z(t)$ is free of overshoot.

Case 3: $b_u(s) = s^2 - 2\zeta\omega_n s + \omega_n^2$ with $\zeta \geq 0$, $\omega_n > 0$, i.e., the plant only has two complex conjugate or real unstable zeros.

In this case, set

$$\alpha_1 = \alpha_2 = \omega_n$$

Then

$$G(s) = \frac{s^2 - 2\zeta\omega_n s + \omega_n^2}{(s + \omega_n)^2} \prod_{i=3}^k \frac{\alpha_i}{s + \alpha_i}.$$

The conclusion in Problem 4.20.3 implies that $z(t)$ is free of overshoot.

In the general case, we cannot simply extend the design ideas above. We need to follow a different idea. Let us set $\alpha_i = \alpha$ for all i . Then the tracking error for a unit step reference is

$$E(s) = [1 - G(s)] \frac{1}{s} = \left[1 - \frac{b_u(s)\alpha^k}{b_u(0)(s + \alpha)^k} \right] \frac{1}{s} = \frac{b_u(0)(s + \alpha)^k - b_u(s)\alpha^k}{b_u(0)(s + \alpha)^k s}.$$

Notice that the numerator is

$$\begin{aligned} & b_u(0)[(s + \alpha)^k - \alpha^k] - [b_u(s) - b_u(0)]\alpha^k \\ &= b_u(0)s[(s + \alpha)^{k-1} + (s + \alpha)^{k-2}\alpha + \cdots + (s + \alpha)\alpha^{k-2} + \alpha^{k-1}] - [b_u(s) - b_u(0)]\alpha^k. \end{aligned}$$

Also notice that the second term above can be divided by s and hence, according to the partial fractional expansion formulas in Section A.3,

$$\frac{b_u(s) - b_u(0)}{b_u(0)(s + \alpha)^k s} = \frac{\beta_1(\alpha)}{s + \alpha} + \frac{\beta_2(\alpha)}{(s + \alpha)^2} + \cdots + \frac{\beta_k(\alpha)}{(s + \alpha)^k}$$

where $\beta_i(\alpha)$, $i = 1, 2, \dots, k$, are polynomial functions of α . This shows that

$$E(s) = \frac{\gamma_1(\alpha)}{s + \alpha} + \frac{\gamma_2(\alpha)}{(s + \alpha)^2} + \cdots + \frac{\gamma_k(\alpha)}{(s + \alpha)^k}$$

where for $i = 1, 2, \dots, k$,

$$\gamma_i(\alpha) = \alpha^{i-1} - \beta_i(\alpha)\alpha^k = \alpha^{i-1}[1 - \beta_i(\alpha)\alpha^{k-i+1}]$$

As long as α is sufficiently small, $\gamma_i(\alpha)$, $i = 1, 2, \dots, k$, are nonnegative. Consequently,

$$e(t) = \mathcal{L}^{-1}[E(s)] = \left(\gamma_1(\alpha) + \gamma_2(\alpha)t + \dots + \frac{\gamma_k(\alpha)}{(k-1)!}t^{k-1} \right) e^{-\alpha t} \sigma(t)$$

is always nonnegative, which means that $z(t)$ has no overshoot

4.8 TIME-DOMAIN SIGNAL AND SYSTEM NORMS

In the analysis and design of control systems, it is often required to measure the strength of a signal or a system. Such a strength, in mathematical terms, is called a **norm**. We often use the norm of a certain signal, such as the tracking error or the plant input (when the reference signal is fixed), or a certain closed-loop system as the performance index of the control system. The purpose in the design is then to find the parameters of the controller so that the value of the performance index is small or minimized. There are several ways to define norms of signals and systems. In this section, we will first look at several time-domain norms. In Chapter 8, we will introduce the frequency-domain norms.

Definition 4.23. Let $x(t)$ be a signal.

1. The ∞ -norm of $x(t)$ is defined as¹

$$\|x(t)\|_{\infty} = \sup_{t \geq 0} |x(t)|.$$

2. The 2-norm of $x(t)$ is defined as

$$\|x(t)\|_2 = \left[\int_{0-}^{\infty} x^2(t) dt \right]^{1/2}.$$

3. The 1-norm of $x(t)$ is defined as

$$\|x(t)\|_1 = \int_{0-}^{\infty} |x(t)| dt.$$

All these norms have physical meanings. The ∞ -norm is the peak magnitude or the amplitude of the signal. The square of the 2-norm is often called the **energy of the signal**. If $x(t)$ is the current or voltage of a resistive load, the square of the 2-norm of $x(t)$ is proportional to the total energy consumption of the load. If $x(t)$ is the velocity of a rigid body, then $\|x(t)\|_2^2$ is proportional to its kinetic energy. If $x(t)$ is the height of a point mass, then $\|x(t)\|_2^2$ is proportional to its potential energy. In the case when $x(t)$ is a certain error signal, then the square of the 2-norm

¹Here “sup” stands for supremum. It means least upper bound.